

Roman Young

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EDUCATION

University of California, San Diego | 3.975 GPA

La Jolla, CA | Expected Grad: 2028

B.S. Bioinformatics, Computer Science Minor

Relevant Coursework: Organic Chemistry (I & II), Genetics, Molecular Biology, Data Analysis for Biologists (R), Advanced Bioinformatics Lab (BWA, minimap2, SAMtools), Discrete Math (I & II), Advanced Data Structures, Statistics in Bioinformatics

EXPERIENCE

La Jolla Institute for Immunology | Bioinformatics Developer

La Jolla, CA | Sept. 2025 - Present

- Authored the indel-aware extension to PEPMatch, advancing it from substitution-only matching to full insertion/deletion alignment via a pigeonhole-seeding and bidirectional-DFS algorithm that guarantees complete recall.
- Benchmarked against common aligners **BLAST**, **DIAMOND**, and **MMseqs2**, achieving **100% recall versus 57 to 92%** for those aligners while running **50 to 100x faster** on all accounts and using ~5x less memory (445MB peak vs. 2.1 to 2.3GB).
- Validated correctness against an independent brute-force oracle across **86M+** exhaustive alignment pairs, backed by an ~89-test pytest suite and CI, and shipped four pull requests merged into the official IEDB/PEPMatch repository.

Salk Institute for Biological Studies | Research Intern

La Jolla, CA | March 2025 - Present

- Contributed to the lab's central investigation of how maternal milk microbiota shapes neonatal gut immune development, using a **control vs. cross-fostered** (CT/CF) mouse model to isolate nursing-derived microbial effects from genetics.
- Executed end-to-end immunophenotyping on CT/CF mice: dissection, isolation of **IELs** and **LPLs** from small and large intestine, multicolor antibody staining, and **flow cytometry** to quantify intestinal T-cell and regulatory T-cell populations.
- Analyzed multiplexed **single-cell RNA-seq** datasets in **R/Seurat** (QC, HTO demultiplexing of pooled multi-mouse samples, clustering) to resolve immune populations; presented findings at lab meetings.

LabReach | Creator

La Jolla, CA | March 2026 - Present

- Built a full-stack **RAG** system (**Next.js**, **TypeScript**, **Neon Postgres** + **pgvector**) that pre-researches **445 research labs across 5** San Diego institutions, cutting a student's per-lab vetting from ~1 hour to seconds before a cold email.
- Designed an extractive, non-generative digest that surfaces only quote-backed evidence, eliminating hallucination, and layered attribution gates that quarantined ~2,200 wrong-author papers to keep the corpus contamination-free.
- Adopted by UCSD STEM clubs to run cold-email workshops, so students can identify and contact research labs at scale.

LEADERSHIP & ACTIVITIES

Great Shape International Dental Project | Clinical Assistant

San Jose, CA | Sept. 2023 - Present

- Selected for international humanitarian missions to deliver dental care to underserved populations across the Caribbean.
- Assisted chairside in the treatment of **220+** patients, supporting a range of clinical procedures including diagnostic radiographs for pathology detection, cleanings, restorative fillings, and extractions to alleviate acute oral pain and infection.
- Led culturally sensitive workshops for **800+** international students, promoting preventive hygiene and long-term dental care.

Eagle Scout Service Project | Project Lead

San Jose, CA | Aug. 2023 - Apr. 2024

- Earned **Eagle Scout** by leading an 8-month mural commission from concept to completion, coordinating **500+ volunteer hours** and maintaining continuous communication with beneficiaries to integrate their feedback into the final design.
- Developed a comprehensive risk-mitigation strategy addressing UV degradation and weather-related wear; collaborated with professional muralists to ensure the mural's long-term structural integrity and positive themes of love and inclusivity.

SKILLS

Technical Skills: Python, TypeScript/Next.js, R, SQL, Bash | Git, CI, HPC/Slurm | algorithm design, performance benchmarking | Flow cytometry, mouse dissection & handling, ELISA / IgA assays, H&E/Alcian Blue histology, PCR, genotyping